MIL-G-36509C
4 August 1978
SUPERSEDING
MIL-G-0036509B(DLA-DM)
18 October 1977 and
MIL-G-36509A
1 May 1973

MILITARY SPECIFICATION

GINGIVAL RETRACTION CORDS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers impregnated gingival retraction cords suitable for use for gingival retraction in dental procedures.

1.2 Classification. Gingival retraction cords shall be of the following types, as specified (see 6.1):

Type I - 4 ply cord.
Type II - 2 ply cord.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 2256 - Test for Breaking Load (Strength) and Elongation of Yarn by the Single-Strand Method.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Applications for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Cotton.

3.1.1.1 Type I. Type I cord shall be 4-ply, spun from bleached, virgin, long staple cotton. Each strand of single ply yarn shall be formed by twisting in a direction opposite to that of the finished cord. The finished cord shall be formed by twisting four lengths of single ply yarn 84 + 3 turns per foot in a direction opposite to that of each single strand. The finished cord shall have a soft, expandable texture without a starch or polish finish.

3.1.1.2 Type II. Type II cord shall be 2-ply, spun from bleached, virgin, long staple cotton. Each strand of single ply yarn shall be formed by twisting in a direction opposite to that of the finished cord. The finished cord shall be formed by twisting two lengths of single ply yarn 24 ± 1 turns per foot in a direction opposite to that of each single strand. The finished cord shall have a soft, expandable texture without a starch or polish finish.

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3.1.2 Racemic epinephrine hydrochloride. The racemic epinephrine hydrochloride shall be of high pharmaceutical grade suitable for the purpose intended.

3.2 Description. The type I and type II cotton cords shall be impregnated with a buffered racemic epinephrine hydrochloride solution using a vacuum or roller press process, and shall be dehydrated to a dry stable state. Type I and type II retraction cords shall be white in color and shall both have a minimum length of 72 inches. Ten yards of type I untreated cotton cord shall weigh a minimum of 2.00 grams (2,268 yards per pound). Ten yards of type II untreated cotton cord shall weigh a minimum of 2.50 grams (1,810 yards per pound).

3.3 Physical properties.

3.3.1 Breaking strength. Type I retraction cord shall have a minimum average breaking strength of 6.0 pounds, when tested in accordance with 4.3.1. Type II retraction cord shall have a minimum average breaking strength of 7.0 pounds, when tested in accordance with 4.3.1.

3.3.2 Epinephrine hydrochloride content. Type I cord shall contain 0.44 to 0.61 milligram (mg) of racemic epinephrine hydrochloride per inch of cord, when tested in accordance with 4.3.2. Type II cord shall contain 0.42 to 0.66 mg of racemic epinephrine hydrochloride per inch of cord, when tested in accordance with 4.3.2.

3.4 Compatibility. The immediate containers and all components shall be compatible and shall not interact physically or chemically with the retraction cords.

3.5 Cleanliness. The immediate containers shall be free from all dirt, foreign matter, or other contaminants, both before and after filling.

3.6 Instructions for use. Adequate and accurate instructions for the use of the retraction cords shall accompany each unit package. The label of each immediate container and unit package shall contain the following legend:

"Caution: Federal law restricts this product to sale and use on the order of a dentist."

3.7 Delivery schedule. Not more than 3 months shall have elapsed from the date of manufacture of the retraction cords to the date of delivery to the Government.

3.8 Lot or control number. Each unit package shall be marked with the lot or control number, and the date of manufacture.

3.9 Workmanship. Workmanship shall be first class throughout. Gingival retraction cords shall be free from defects which detract from their appearance or impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw materials, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Examination. The retraction cords shall be examined for defects including, but not limited to, those listed in table I.

TABLE I. Classification of defects.

Categories	Defects
Major:	
101	Type not as specified.
102	Color not as specified.
103	Immediate containers not clean. 1/
104	Texture of cords not as specified.
105	Instructions not adequate or legible.
Minor:	
151	Length not as specified for specific type. 1/
152	Weight of untreated cotton cords not as specified.

1/ In process examination may be employed.

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The inspection levels and acceptable quality levels (AQL's) shall be as listed in table II. The unit of product for examination purposes shall be one package.

TABLE II. Examination criteria.

Type defect	Inspection level	AQL (Percent defective)
Major	S-2	1.5
Minor	S-2	4.0

4.2.2 Testing. Cords shall be tested as specified in table III and 4.3. A sufficient quantity of retraction cord from each lot or control number shall be selected for testing.

TABLE III. Test criteria.

Characteristic	Requirement	Test procedure	Number of determinations per lot
Breaking strength	3.3.1	4.3.1	5 1/
Assay of epinephrine hydrochloride	3.3.2	4.3.2	3 2/

1/ The lot shall be rejected if the average of the determinations fails the test requirement.
2/ Three determinations shall be made on one sample. The lot shall be rejected if the average of the determinations fails the test requirement.

4.2.2.1 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.2.3 Packaging inspection. The inspection of the unit and intermediate package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.3 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.3.1 Breaking strength. The breaking strength of the retraction cords shall be determined by American Society for Testing and Materials (ASTM) method Designation: D 2256, except that five samples from each lot or control number shall be tested and the results averaged.

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4.3.2 Assay.

4.3.2.1 Test reagents. Dissolve 1.5 grams of ferrous sulfate heptahydrate in 200 milliliters (ml) of distilled water to which has been added 1 ml of 1 N hydrochloric acid and 1.0 gram of sodium bisulfite. The amount of iron in 0.1 ml of this solution is sufficient for the determination of quantities of epinephrine up to 0.5 mg. For larger amounts of epinephrine, it is necessary to employ a proportionately higher concentration of ferrous sulfate. Prepare the ferrous sulfate-citrate solution by adding 0.5 gram of sodium citrate ($Na_3C_6H_5O_7.2H_2O$) to 10 ml of the first solution. This solution should be prepared fresh each day. To prepare the buffer reagent, add 42 grams of sodium bicarbonate and 50 grams of potassium bicarbonate to 180 ml of distilled water. Not all of the solids will be dissolved at this stage. To another 180 ml of water, add 37.5 grams of aminoacetic acid and 17 ml of strong ammonia solution (28 percent NH_3). Mix the two solutions and add sufficient water to adjust the volume to 500 ml. The remaining solids will be dissolved at usual room temperatures after the final addition of water.

4.3.2.2 Assay test. Suspend a 5-inch length of retraction cord in 50 ml of distilled water and allow to soak at room temperature for 15 minutes. Remove a 10 ml aliquot with a pipet, place in a test tube and add 0.1 ml of ferrous-citrate reagent, followed by 1.0 ml of the buffer reagent. Mix the resulting solution and allow to stand for at least 10 minutes to develop full color. The color developed is read in a spectrophotometer at a wave length of 530 millimicrons. The concentration of racemic epinephrine hydrochloride is read directly from a calibration curve. The calibration curve is prepared by plotting optical density against the concentration of racemic epinephrine hydrochloride. The density readings are obtained after the instrument has been set to read 0.0 when the comparison tube contains distilled water. Three determinations shall be conducted and the results shall be averaged.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One package containing 72 inches of type I or type II cord constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit packaging. Seventy-two inches of type I or type II cord shall be packaged in a plastic dispenser type container having a suitable closure to prevent contamination or deterioration of the cord. Each dispenser along with instructions for use shall be further packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure of the box shall be as specified in the appendix of the applicable box specification.

5.1.2.2 Intermediate package. Twelve unit packages containing cord of one type only shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

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5.1.3 Level C. Seventy-two inches of type I or type II cord shall be packaged in a dispenser as specified in 5.1.2.1. Each dispenser along with instructions for use shall be further packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Forty-eight intermediate packages (576 units) containing type I or type II cord shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping, and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Forty-eight intermediate packages (576 units) containing type I or type II cord shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged cord shall be packed in shipping containers in a manner that will afford adequate protection against damage during shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Dispenser. Each dispenser shall be labeled in accordance with the requirements of the Federal Food, Drug, and Cosmetic Act and the contractor's commercial practice. The lot or control number shall be shown.

5.3.2 Interior packages. Each unit and intermediate package shall be marked as specified in MIL-STD-129. The lot or control number, contract or purchase order number, and name of contractor shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)".

5.3.3 Shipping container. Each shipping container shall be marked as specified in MIL-STD-129. The lot or control number and the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)" shall be shown.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 Epinephrine hydrochloride assay. The method of assay used for racemic epinephrine hydrochloride content is the J.R. Doty method as described in Analytical Chemistry 20, 1166, 1948.

6.3 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

NSN	Item Identification	Specification Designation
6520-00-890-1864	GINGIVAL RETRACTION CORD, IMPREGNATED, DENTAL, 4 Ply.	Type I
6520-00-890-1865	GINGIVAL RETRACTION CORD, IMPREGNATED, DENTAL, 2 Ply.	Type II

6.4 Changes form previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA-DM

Project No. 6520-1472

Review activity:
DoD-MB

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-G-36509C GINGIVAL RETRACTION CORDS

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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